

COM S 4740/5740: Introduction to Machine Learning

Homework #1

1. Please submit your report as “HW#_FirstName_LastName.pdf” on Canvas. The report should be in PDF format and include your answers to all questions. You are required to include figures in your report.
 2. Unlimited number of submissions are allowed on Canvas and the latest one will be graded.
 3. **No later submission is accepted.**
 4. Please read and follow submission instructions. No exception will be made to accommodate incorrectly submitted files/reports.
 5. All students are required to typeset their reports using latex. Overleaf (<https://www.overleaf.com/learn/latex/Tutorials>) can be a good start.
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1. (10 points) Consider the perceptron in two dimensions: $h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x})$ where $\mathbf{w} = [w_0, w_1, w_2]^T$ and $\mathbf{x} = [1, x_1, x_2]^T$. Technically, $\mathbf{x}$ has three coordinates, but we call this perceptron two-dimensional because the first coordinate is fixed at 1.
 - (a) Show that the regions on the plane where $h(\mathbf{x}) = +1$ and $h(\mathbf{x}) = -1$ are separated by a line. If we express this line by the equation $x_2 = ax_1 + b$, what are the slope a and intercept b in terms of w_0, w_1, w_2 ?
 - (b) Draw pictures for the cases $\mathbf{w} = [1, 2, 3]^T$ and $\mathbf{w} = -[1, 2, 3]^T$. Label the positive and negative prediction regions on the pictures.
2. (30 points) In logistic regression (labels are $\{1, -1\}$), the objective function can be written as

$$E(w) = \frac{1}{N} \sum_{n=1}^N \ln \left(1 + e^{-y_n w^T x_n} \right).$$

Please

- (a) (10 points) Compute the first-order derivative $\nabla E(w)$. You will need to provide the intermediate steps of derivation.
- (b) (10 points) Once the optimal w is obtain, it will be used to make predictions as follows:

$$\text{Predicted class of } x = \begin{cases} 1 & \text{if } \theta(w^T x) \geq 0.5 \\ -1 & \text{if } \theta(w^T x) < 0.5 \end{cases}$$

where $\theta(z) = \frac{1}{1+e^{-z}}$.

Explain why the decision boundary of logistic regression is still linear, though the linear signal $w^T x$ is passed through a nonlinear function θ to compute the outcome of prediction.

- (c) (5 points) Is the decision boundary still linear if the prediction rule is changed to the following? Justify briefly.

$$\text{Predicted class of } x = \begin{cases} 1 & \text{if } \theta(w^T x) \geq 0.9 \\ -1 & \text{if } \theta(w^T x) < 0.9 \end{cases}$$

- (d) (5 points) In light of your answers to the above two questions, what is the essential property of logistic regression that results in the linear decision boundary?

3. (10 points) Given

$$X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{m \times n}$$

where $x_i \in \mathbb{R}^m$ for all i , and

$$Y = \begin{bmatrix} y_1^T \\ y_2^T \\ \vdots \\ y_n^T \end{bmatrix} \in \mathbb{R}^{n \times p}$$

where $y_i \in \mathbb{R}^p$ for all i . Show that

$$XY = \sum_{i=1}^n x_i y_i^T.$$

4. (10 points) Show that sigmoid function and softmax function are the same in the binary case.

$$\begin{aligned} \text{sigmoid}(y) &= \frac{1}{1 + e^{-y}} \\ \text{softmax}(y_j) &= \frac{e^{y_j}}{\sum_{i=1}^c e^{y_i}} \end{aligned}$$